

PROJECT TITLE: OptiCrop: Smart Agricultural Production Optimization Engine

1. APPLICATION CORE INTEGRATION ARCHITECTURE STACK:

The system employs a standardized modular data-driven programming architecture split cleanly between structural presentation models and mathematical predictive computing routines:

- * Programming Language Environment: Python 3.13 Standard Engine Configuration
- * Analytical Matrix Computation Core: NumPy Packages Core Layout Architecture
- * Tabular Frame Data Manipulation: Pandas DataFrame Processing Library
- * Machine Learning Optimization Framework: Scikit-learn (Random Forest Ensemble Estimators)
- * Web Platform Integration Framework: Flask (Python Micro Web Framework Engine Architecture)
- * Frontend Interface Presentation: HTML5 Layout Framework integrated with Bootstrap v5 Responsive Design Component Mapping